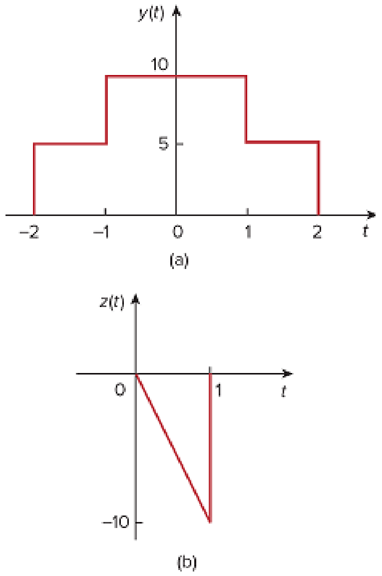
Chapter 18, Problem 9P

Problem

Determine the Fourier transforms of the signals in Fig. 18.34.

Figure 18.34



Step-by-step solution

Step 1 of 4

(a)

Refer to signal in Figure 18.34 (a) in the textbook.

The signal can be expressed as,

$$y(t)=\begin{cases}5 & -2\le t\le -1\\ 10 & -1\le t\le 0\\ 10 & 0\le t\le 1\\ 5 & 1\le t\le 2\end{cases}$$

Write the expression for the Fourier transform.

$$F(\omega)=\int_{-\infty}^{\infty} f(t)e^{-j\omega t}\,dt$$

Step 2 of 4

Calculate the Fourier transform.

$$Y(\omega)=\int_{-\infty}^{\infty} y(t)e^{-j\omega t}\,dt$$

$$=\int_{-2}^{-1} 5e^{-j\omega t}\,dt+\int_{-1}^{0} 10e^{-j\omega t}\,dt+\int_{0}^{1} 10e^{-j\omega t}\,dt+\int_{1}^{2} 5e^{-j\omega t}\,dt$$

$$=\int_{-2}^{-1} 5e^{-j\omega t}\,dt+\int_{-1}^{0} 10e^{-j\omega t}\,dt+\int_{0}^{1} 10e^{-j\omega t}\,dt+\int_{1}^{2} 5e^{-j\omega t}\,dt$$

$$=5\left[\frac{e^{-j\omega t}}{-j\omega}\right]_{-2}^{-1}+10\left[\frac{e^{-j\omega t}}{-j\omega}\right]_{-1}^{0}+10\left[\frac{e^{-j\omega t}}{-j\omega}\right]_{0}^{1}+5\left[\frac{e^{-j\omega t}}{-j\omega}\right]_{1}^{2}$$

$$=\frac{5}{j\omega}\left[e^{j2\omega}-e^{j\omega}\right]+\frac{10}{j\omega}\left[e^{j\omega}-e^{0}\right]+\frac{10}{j\omega}\left[e^{0}-e^{-j\omega}\right]+\frac{5}{j\omega}\left[e^{-j\omega}-e^{-j2\omega}\right]$$

$$=\frac{5}{j\omega}\left[e^{j2\omega}-e^{j\omega}\right]+\frac{10}{j\omega}\left[e^{j\omega}-1\right]+\frac{10}{j\omega}\left[1-e^{-j\omega}\right]+\frac{5}{j\omega}\left[e^{-j\omega}-e^{-j2\omega}\right]$$

$$=\frac{5}{j\omega}\left[e^{j2\omega}-e^{-j2\omega}\right]+\frac{10}{j\omega}\left[e^{j\omega}-e^{-j\omega}\right]+\frac{10}{j\omega}\left[1-1\right]+\frac{5}{j\omega}\left[e^{-j\omega}-e^{j\omega}\right]$$

$$=\frac{2\times 5}{j\omega}\left[\frac{e^{j2\omega}-e^{-j2\omega}}{2}\right]+\frac{2\times 10}{j\omega}\left[\frac{e^{j\omega}-e^{-j\omega}}{2}\right]+0-\frac{2\times 5}{j\omega}\left[\frac{e^{j\omega}-e^{-j\omega}}{2}\right]$$

Step 3 of 4

On further simplification:

$$Y(\omega)=\frac{10}{\omega}\left[\frac{e^{j2\omega}-e^{-j2\omega}}{j2}\right]+\frac{20}{\omega}\left[\frac{e^{j\omega}-e^{-j\omega}}{j2}\right]+0-\frac{10}{\omega}\left[\frac{e^{j\omega}-e^{-j\omega}}{j2}\right]$$

$$=\frac{10}{\omega}\left[\frac{e^{j2\omega}-e^{-j2\omega}}{j2}\right]+\frac{20}{\omega}\left[\frac{e^{j\omega}-e^{-j\omega}}{j2}\right]+-\frac{10}{\omega}\left[\frac{e^{j\omega}-e^{-j\omega}}{j2}\right]$$

$$=\frac{10}{\omega}\sin 2\omega+\frac{20}{\omega}\sin\omega-\frac{10}{\omega}\sin\omega$$

$$=\frac{10}{\omega}\sin 2\omega+\frac{\sin\omega}{\omega}(20-10)$$

$$=\frac{10}{\omega}\sin 2\omega+10\frac{\sin\omega}{\omega}$$

$$=\frac{10}{\omega}\sin 2\omega+\frac{10}{\omega}\sin\omega$$

Therefore, the Fourier transform, $Y(\omega)$ is $\boxed{\frac{10}{\omega}\sin 2\omega+\frac{10}{\omega}\sin\omega}$.

Step 4 of 4

(b)

Refer to signal in Figure 18.34 (a) in the textbook.

The signal can be expressed as,

$$y(t) = -10t \qquad 0 \le t \le 1$$

Calculate the Fourier transform.

$$Y(\omega) = \int_{-\infty}^{\infty} y(t) e^{-j\omega t} dt$$

$$= \int_{0}^{1} -10te^{-j\omega t} dt$$

$$= -10\left[\frac{e^{-j\omega t}}{(-j\omega)^2}(-j\omega t - 1)\right]_0^1 \qquad \left(\text{since } \int e^{ax} dx = \frac{e^{ax}}{a^2}(ax-1)+C\right)$$

$$= -10\left[\frac{e^{-j\omega}}{(-j\omega)^2}(-j\omega - 1) - \frac{e^0}{(-j\omega)^2}(-j\omega(0)-1)\right]$$

$$= 10\left[\frac{e^{-j\omega}}{\omega^2}(-j\omega-1) + \frac{1}{\omega^2}\right]$$

$$= \frac{10}{\omega^2} - \frac{10e^{-j\omega}}{\omega^2}(1+j\omega)$$

Therefore, the Fourier transform, $Y(\omega)$ is $\boxed{\frac{10}{\omega^2} - \frac{10e^{-j\omega}}{\omega^2}(1+j\omega)}$.